

T-Test

Group Members

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Subject: **Tools and Methods of Data Analysis**

Date: **21/05/2026**

1. Introduction

This report summarizes our statistical toolkit in Python and how to use it to perform and interpret three One-sample, independent two-samples and paired t-tests. The primary underlying principle of all the tests is — Recognizing a true pattern in data or if it is a chance pattern. We played with three real-world Datasets on student sleep, academic grades, and personal health metrics from Kaggle.

2. Data Description

Table 1: Datasets used in this analysis

Test	File	n	Column(s) Used
One-Sample	student_stress_sleep_screen.csv	150	sleep_hours
Two-Sample	student-mat.csv	395	G3 by school (GP vs. MS)
Paired	Health_Sleep_Statistics.csv	100	Daily Steps vs. Calories Burned

3. Data Visualisation

3.1. One-Sample t-Test: Student Sleep Hours

Figure 1 — One-Sample t-Test: Student Sleep Hours

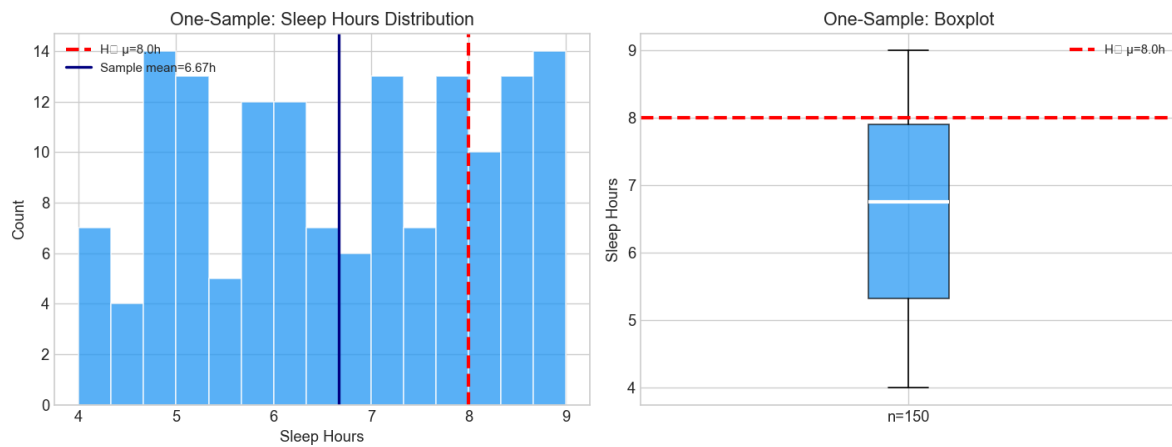


Figure 1: Histogram and boxplot of sleep hours. Red dashed line: $\mu_0 = 8h$. Navy line: sample mean.

3.2. Two-Sample t-Test: Final Grades by School

Figure 2 — Two-Sample t-Test: GP vs MS Final Grades

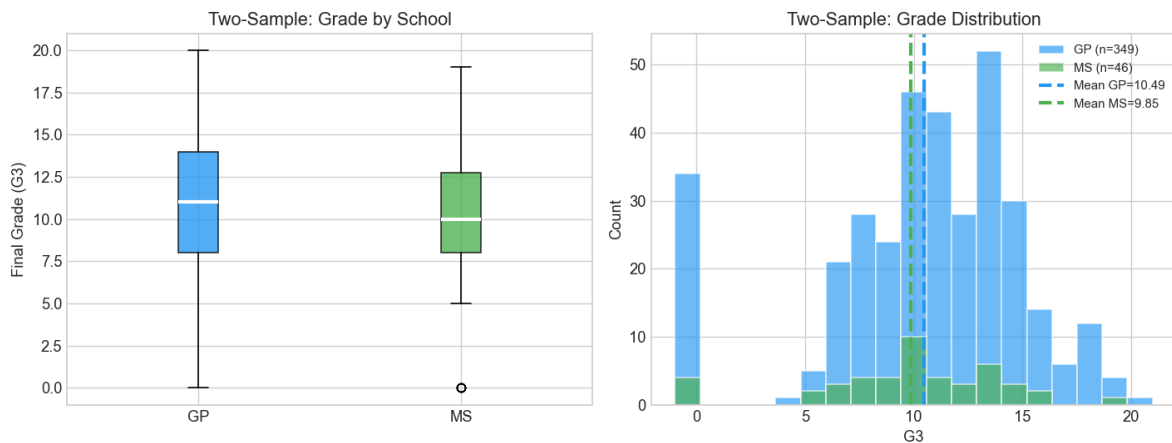


Figure 2: Boxplots and histograms for G3 grades: GP vs. MS. Dashed lines show group means.

3.3. Paired t-Test: Daily Steps vs. Calories Burned

Figure 3 — Paired t-Test: Daily Steps vs Calories Burned

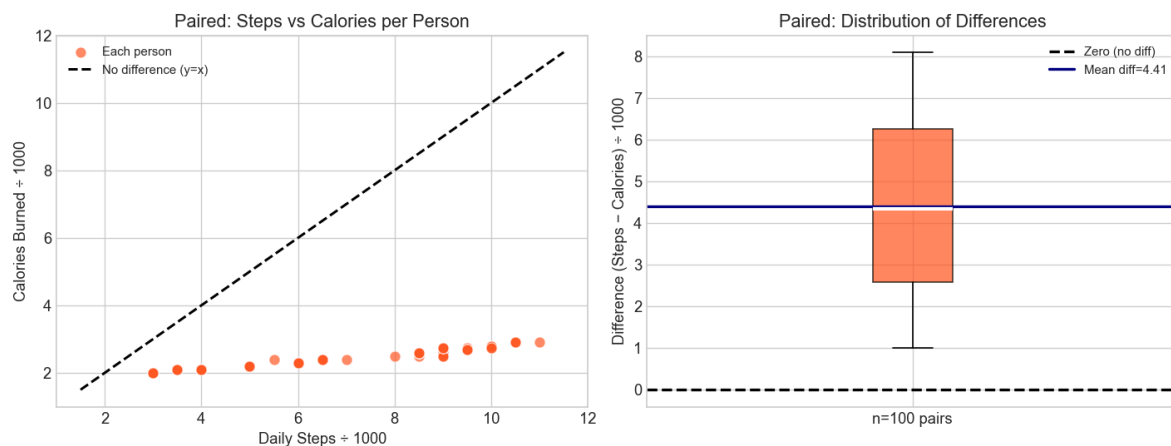


Figure 3: Scatter plot of paired observations and boxplot of within-person differences. Dashed diagonal: $y = x$.

4. Assumption Checks

Table 2: Assumption checks (fill values from notebook Cells 6, 8, 10)

Test	Check	Stat	p-value	Met?
One-Sample	Shapiro-Wilk (normality)			
Two-Sample	Shapiro-Wilk GP / MS			
Two-Sample	Levene's test (equal variances)			
Paired	Shapiro-Wilk on differences			

If Levene's test gives $p < 0.05$, Welch's correction is applied automatically via `equal_var=False` in our toolkit.

5. Test Results and Confidence Intervals

All three tests use $\alpha = 0.05$. Fill in the blank cells below using the printed output from the Jupyter Notebook.

5.1. One-Sample t-Test

$$H_0 : \mu = 8.0 \text{ h} \quad H_a : \mu \neq 8.0 \text{ h}$$

Parameter	Value	Parameter	Value
n		t -statistic	
$\bar{x}$		df	
μ_0	8.0 h	p -value	
Mean difference		95% CI	
Decision			

Table 3: One-Sample t-Test results

5.2. Independent Two-Sample t-Test (Welch)

$$H_0 : \mu_{GP} = \mu_{MS} \quad H_a : \mu_{GP} \neq \mu_{MS}$$

Parameter	Value	Parameter	Value
n_{GP} / n_{MS}		t -statistic (Welch)	
Mean GP / MS		df (Welch)	
Mean difference		p -value	
95% CI		Decision	

Table 4: Two-Sample t-Test results

5.3. Paired t-Test

$$H_0 : \mu_d = 0 \quad H_a : \mu_d \neq 0$$

Parameter	Value	Parameter	Value
n (pairs)		t -statistic	
Mean diff $\bar{d}$		df	
95% CI		p -value	
Decision			

Table 5: Paired t-Test results

6. Conclusions

One-Sample: We wanted to know if students get the 8 hours of sleep they are supposed to have per night (One-Sample). The test returned $t(df) = \text{value}$, $p = \text{value}$. Since p [is/is not] ≤ 0.05 , we [reject/fail to reject] H_0 , indi- Sleeping for 8 hours [significantly less/more/no differently from] students who do not sleep for 8 hours. The 95[excludes/includes] 0 = 8.

Two-Sample: Comparing GP and MS final grades, the Welch test was used, $t(df) = \text{value}$, $p = \text{value}$. We [reject/fail to reject] H_0 . The mean difference of [value] grade points (95% CI [. . . , . . .]) [does/does not] indicate anything but a rather unhelpful difference between the two schools.

Paired: Testing within-person differences between steps and calories burned, the paired test returned $t(df) = \text{value}$, $p = \text{value}$. We [reject/fail to reject] H_0 . The paired design was appropriate here as it removes between-person variability, giving a cleaner test of the two measurements per individual.

7. Toolkit and Source Code

The analysis was produced using an 11-cell Jupyter Notebook with `numpy`, `pandas`, `matplotlib`, and `scipy.stats`. Each cell includes comments on what it does and which parameters to change for different results.

Source code: <https://github.com/PathakMan/T-Test>

References

1. Bhaskar, A. (2024). *Student Stress, Sleep and Screen Time Dataset*. Kaggle. <https://www.kaggle.com/datasets/arpitabhaskar/student-stress-sleep-and-screen-time-dataset>
2. Cortez, P. (2014). *Student Performance Data Set*. UCI / Kaggle. <https://www.kaggle.com/datasets/dskagglemt/student-performance-data-set>
3. Aksoy, H. (2023). *Health and Sleep Statistics*. Kaggle. <https://www.kaggle.com/datasets/hanaksoy/health-and-sleep-statistics>